Thermal-Aware Power Budgeting on a Silicon Die: A PDE-Constrained Linear Programming Approach

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Abstract

Uneven heat generation on a processor die creates local hotspots, and sustained high temperature accelerates electromigration and device ageing. This work asks how a fixed budget of 5 W should be divided among four heat-dissipating blocks on a 10 mm × 10 mm silicon die so that the highest temperature anywhere on the die is as low as possible. Steady conduction is described by the two-dimensional Poisson equation with isothermal edges at 300 K, discretised with a five-point finite-difference stencil on a 40×40 grid. Inverting the resulting sparse system yields a thermal impedance matrix, and the minimax allocation is then written as a linear program through an epigraph reformulation and solved with HiGHS. The discrete heat equation is solved to a relative residual below 10^{-13}, and the full 1605-variable program and a reduced five-variable program agree to within 10^{-8} K. Two findings follow. With four identical point-like blocks in a symmetric layout the equal split is already optimal. When the blocks have footprints of 0.5, 1.0, 1.5 and 2.0 mm, the optimal split gives the smallest block 0.90 W and the largest 1.57 W, lowers the peak from 309.15 K to 307.04 K (a 23.1% cut in the temperature rise above ambient) and equalises the peak temperature of all four blocks. The formulation also exposes a modelling pitfall: if the budget is written as an inequality, zero power is trivially optimal, so it must be imposed as an equality.

Keywords: thermal management; Poisson equation; finite differences; thermal impedance; linear programming; minimax optimisation

1. Introduction

Modern processors place several functional blocks, such as cores, caches and accelerators, within a few square millimetres of silicon, and each block turns electrical power into heat. When one block dissipates much more power per unit area than its neighbours, its temperature rises well above the die average. This matters beyond comfort, because current-driven atom transport in metal interconnect, known as electromigration, speeds up sharply with temperature [9], and the expected lifetime of the chip falls accordingly. The way a total power budget is shared between blocks is one of the few levers that a designer or a runtime power manager can pull without changing the hardware, so it deserves a principled answer.

The problem studied here is as follows. A total power budget $P_{\max} = 5$ W is to be distributed among $N = 4$ blocks of a square silicon die, with block powers $q_1, \dots, q_4 \geq 0$ that sum to the budget, so that the peak temperature over the whole die is minimised. The die edges are held at $T_{\mathrm{amb}} = 300$ K by an ideal heat sink. The answer is not obvious by inspection, because every temperature depends on all four powers at once and because the hottest point can move from one block to another as the powers change.

We solve the problem in three connected steps. The governing equation is first derived from Fourier's law and energy conservation. It is then discretised by finite differences, which turns the temperature field into a sparse linear system and, after inversion, into a thermal impedance matrix giving the temperature rise at every point per watt injected into every block. Finally we observe that the peak temperature is a convex, piecewise-linear function of the block powers, so that its minimisation is a linear program with an exactly verifiable optimum. The remainder of the report reviews related work (Section 2), develops the mathematics (Section 3), describes the computational set-up (Section 4), interprets the results (Section 5), states the limitations (Section 6) and concludes with future directions (Section 7).

2. Related Work

Heat conduction in solids is treated in standard textbooks such as Incropera et al. [1], which give Fourier's law and the steady heat equation used here. The finite-difference treatment of the resulting elliptic problem, including the five-point Laplacian and its linear-algebra properties, is set out by LeVeque [2], and the Laplace-transform route from the transient equation to a frequency-domain impedance is the standard engineering-mathematics treatment found in Kreyszig [3]. In the architecture literature, Skadron et al. showed that on-chip temperature can be modelled at the granularity of functional blocks and managed dynamically, for instance by throttling activity once a threshold is crossed [4]. Their HotSpot methodology represents die, package and heat sink as a compact network of thermal resistances and capacitances, analogous to an electrical circuit [5]. On the optimisation side, minimax problems and their epigraph form as linear programs are textbook material [6], and the HiGHS solver used here implements a modern dual simplex method [7].

These ingredients are usually applied separately. Compact thermal models mainly analyse or react to temperature, and dynamic thermal management policies are generally heuristic and carry no certificate that a better

power split does not exist. The gap addressed here is the combination of a field-level conduction model with an exact convex optimisation, so that the best static power split for a given floorplan is computed directly and its optimality can be checked. We do not claim that this combination is new; the contribution is a transparent and reproducible implementation, a check of its numerical soundness, and an honest account of when the optimiser does and does not help.

3. Methodology

3.1 Field model and governing equation

Heat flows from hot to cold at a rate proportional to the local temperature gradient, which Fourier's law expresses as

$$\mathbf{q}''(x,y) = -k\,\nabla T(x,y) = -k\Big(\frac{\partial T}{\partial x}\,\hat{\imath} + \frac{\partial T}{\partial y}\,\hat{\jmath}\Big). \tag{1}$$

Here $\mathbf{q}''$ is the heat-flux vector ($\mathrm{W\,m^{-2}}$), T the temperature (K), $k = 150\ \mathrm{W\,m^{-1}K^{-1}}$ the thermal conductivity of silicon, and $\hat{\imath}, \hat{\jmath}$ the in-plane unit vectors; the minus sign records that heat flows down the gradient. The flux is written $\mathbf{q}''$ to keep it distinct from the block powers q_j used later. In a steady state the net heat leaving any small region equals the heat generated in it, and by the divergence theorem this local balance reads

$$\nabla\cdot\,\mathbf{q}'' = Q(x,y), \tag{2}$$

where Q ($\mathrm{W\,m^{-3}}$) is the volumetric heat generation, nonzero only inside a functional block. Substituting Eq. (1) into Eq. (2) with constant k gives the two-dimensional Poisson equation

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = -\frac{Q(x,y)}{k}, \qquad T\big|_{\partial\Omega} = T_{\mathrm{amb}} = 300\ \mathrm{K}, \tag{3}$$

in which ∇^2 is the Laplacian, Ω the square die and $\partial\Omega$ its boundary. The equation is elliptic, so for a given Q it has exactly one smooth solution. The Dirichlet condition fixes the boundary at ambient and represents an ideal heat sink on all four edges. It is convenient to work with the temperature rise $\theta = T - T_{\mathrm{amb}}$, which obeys the same equation with $\theta = 0$ on $\partial\Omega$.

3.2 Transient behaviour and thermal impedance

Before the steady state is reached the die obeys $\rho c_p\,\partial\theta/\partial t = k\nabla^2\theta + Q$, where ρ is the density, c_p the specific heat and t time, starting from $\theta(\cdot,0) = 0$. The Laplace transform in time, with complex frequency s and bars denoting transformed quantities, replaces the time derivative by multiplication by s:

$$k\nabla^2\bar{\theta}(\mathbf{r},s) - \rho c_p\,s\,\bar{\theta}(\mathbf{r},s) = -\bar{Q}(\mathbf{r},s). \tag{4}$$

The ratio of the transformed temperature rise to the transformed power is the thermal impedance $Z_{th}(s)$. For a power step of size P, whose transform is P/s, the final-value theorem gives $\theta(\infty) = \lim_{s\to 0} s\bar{\theta} = Z_{th}(0)\,P$. As $s \to 0$ the term $\rho c_p s\bar{\theta}$ in Eq. (4) vanishes and the equation collapses to Eq. (3), so the zero-frequency impedance is exactly the steady-state temperature rise per watt. A single-node check makes this concrete. For a thermal resistance R in parallel with a heat capacity C, the transformed balance $Cs\bar{\theta} = -\bar{\theta}/R + \bar{P}$ gives $Z_{th}(s) = R/(1+RCs)$, whose limit at $s \to 0$ is R; SymPy confirms this and returns the familiar step response $R(1 - e^{-t/RC})$. For the full die the same limit is evaluated numerically by the discretisation below.

3.3 Finite-difference discretisation

The die is covered by a uniform grid of $40 \times 40 = 1600$ nodes with spacing $\Delta x = \Delta y = 0.25$ mm. Writing $\theta_{i,j}$ for the rise at node (i,j), replacing the second derivatives in Eq. (3) by central differences gives the five-point stencil

$$\frac{\theta_{i+1,j} + \theta_{i-1,j} + \theta_{i,j+1} + \theta_{i,j-1} - 4\theta_{i,j}}{\Delta x^2} = -\frac{Q_{i,j}}{k}, \tag{5}$$

so that each interior node is coupled to its four neighbours; nodes on the outer ring are assigned $\theta = 0$. Numbering the nodes by $n = 40i + j$ and stacking the equations gives the sparse system

$$\mathbf{K}\,\boldsymbol{\theta} = \mathbf{B}\,\mathbf{q}. \tag{6}$$

In this equation $\boldsymbol{\theta} \in \mathbb{R}^{1600}$ holds the nodal rises, $\mathbf{q} = (q_1, \ldots, q_4)^{\top}$ the block powers in watts, and $\mathbf{K} \in \mathbb{R}^{1600\times 1600}$ is the negative discrete Laplacian: an interior row has $4/\Delta x^2$ on the diagonal and $-1/\Delta x^2$ in the four neighbour columns, and a boundary row is an identity row that enforces $\theta = 0$. The negative sign is used so that $\mathbf{K}$ is positive definite and all impedances come out positive. The matrix $\mathbf{B} \in \mathbb{R}^{1600\times 4}$ turns block power into the source term of Eq. (5). If block b covers n_b nodes and the die has thickness t_d, its power is spread uniformly over that footprint and through the thickness, so $Q_{i,j} = q_b/(n_b \Delta x^2 t_d)$ inside the block, and therefore $B_{n,b} = 1/(n_b \Delta x^2 t_d k)$ for nodes n in block b and zero elsewhere.

Three properties of $\mathbf{K}$ matter for the optimisation. Its interior rows are weakly diagonally dominant, since the diagonal $4/\Delta x^2$ equals the sum of the four neighbour magnitudes, and its boundary rows are strictly dominant, so the matrix is irreducibly diagonally dominant and hence nonsingular. Its off-diagonal entries are non-positive, which makes it an M-matrix and guarantees that $\mathbf{K}^{-1}$ has no negative entries. Solving Eq. (6) for the four unit powers then yields the steady-state thermal impedance matrix

$$\mathbf{Z} = \mathbf{K}^{-1}\mathbf{B} \in \mathbb{R}^{1600\times 4}, \qquad \boldsymbol{\theta} = \mathbf{Z}\mathbf{q}, \tag{7}$$

where $Z_{n,b} \geq 0$ is the temperature rise in kelvin at node n per watt in block b. This is the discrete counterpart of $Z_{th}(0)$, and superposition, which makes the whole problem linear, follows from the linearity of Eq. (6).

3.4 Minimax power allocation as a linear program

The quantity to minimise is the peak rise $\theta_{\max}(\mathbf{q}) = \max_n(\mathbf{Z}\mathbf{q})_n$. A maximum of linear functions is convex and piecewise linear but not itself a linear objective, so an epigraph variable t is introduced that must lie above every nodal rise:

$$\begin{aligned} &\min_{\mathbf{q},\boldsymbol{\theta},t} \; t \\ &\text{s.t. } \mathbf{K}\boldsymbol{\theta} - \mathbf{B}\mathbf{q} = \mathbf{0}, \quad \theta_n \leq t \;\; (n = 1,\ldots,1600), \\ &\qquad q_1 + q_2 + q_3 + q_4 = P_{\max}, \quad q_j \geq 0. \end{aligned} \tag{8}$$

The first constraint is the discretised heat equation, the second makes t an upper bound on the rise at every node, the third spends the whole budget, and the last forbids negative power. At the optimum t equals the smallest achievable peak rise, so the optimal peak temperature is $T_{\text{amb}} + t^\star$. Substituting $\boldsymbol{\theta} = \mathbf{Z}\mathbf{q}$ removes the 1600 temperature variables and leaves a five-variable program, $\min t$ subject to $\mathbf{Z}\mathbf{q} \leq t\mathbf{1}$, $\mathbf{1}^\top\mathbf{q} = P_{\max}$, $\mathbf{q} \geq 0$; both forms are solved and compared as a consistency check.

The budget is written as an equality, and this needs justification because a budget is more naturally an inequality $\sum_j q_j \leq P_{\max}$. Since every entry of $\mathbf{Z}$ is non-negative, lowering any block power can only lower temperatures, so with the inequality the optimiser sets all powers to zero and reports $t^\star = 0$, which is feasible but meaningless. The equality encodes the fact that the workload must actually dissipate $P_{\max}$ watts and leaves only the question of where. A second structural fact helps interpret the results. If the layout is mirror-symmetric, so that reflections of the die permute the blocks and leave $\mathbf{K}$ unchanged, then averaging any optimal $\mathbf{q}$ over those reflections gives an allocation whose peak is no larger, by convexity of the maximum. For four identical blocks in a symmetric layout the equal split is therefore optimal, and the optimiser can only help when the blocks differ.

3.5 Key metrics and pipeline summary

An allocation is judged by its peak temperature $T_{\text{amb}} + \theta_{\max}$, by the reduction $\Delta\theta = \theta_{\max}^{\text{equal}} - \theta_{\max}^{\text{LP}}$ against the equal-power baseline, and by the spread of block peak temperatures, since a minimax optimum equalises the blocks that receive power. Numerical soundness is judged by the relative residual $\|\mathbf{K}\boldsymbol{\theta} - \mathbf{B}\mathbf{q}\|_2 / \|\mathbf{B}\mathbf{q}\|_2$, the spectral condition number of the interior part of $\mathbf{K}$, the solver status, and the agreement of the full and reduced programs. Table 1 records which stage of the pipeline rests on which course.

Table 1: Mathematical pipeline and the course supplying each stage.

Stage	Concept	Course	Implementation here
1. Field model	gradient, divergence	Vector Calculus	Eqs. (1)–(2)
2. PDE	elliptic Poisson, Dirichlet BC	PDEs	Eq. (3)
3. Transform	Laplace in t, $s \to 0$	Transforms	Eq. (4), SymPy check
4. Discretisation	sparse linear system	Linear Algebra	Eqs. (5)–(7), SciPy sparse LU
5. Optimisation	epigraph LP	Optimisation	Eq. (8), HiGHS
6. Verification	residuals, heat maps	Python Lab	Section 5

4. Experimental Setup

All computations were done in Python 3. NumPy 2.4.4 supplies the arrays and mesh bookkeeping, and SciPy 1.17.1 [8] supplies the sparse matrices used to assemble $\mathbf{K}$ (list-of-lists format converted to compressed sparse row), the sparse LU factorisation used to solve Eq. (6) for each column of $\mathbf{B}$, and the `linprog` interface to the HiGHS solver [7]. SymPy 1.14.0 performed the symbolic Laplace-transform check and Matplotlib 3.10.8 produced the figures. A single script regenerates every table and figure; it ran on one CPU core, and a full linear program with 1605 variables solves in about 0.3 s, so no special hardware was needed.

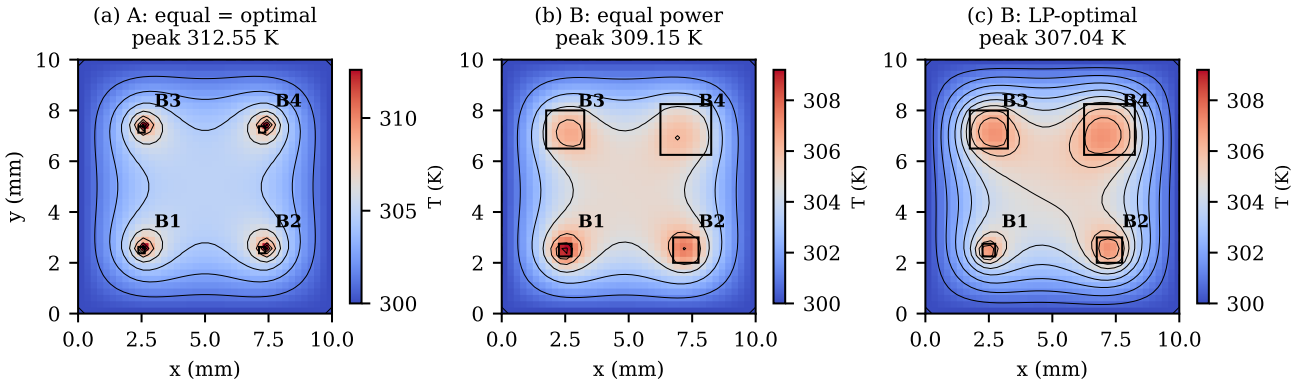


Figure 1: Steady-state temperature fields. (a) Scenario A under equal power, which is also the LP optimum. (b) Scenario B under equal power, where the small block B1 sets the peak. (c) Scenario B under the LP-optimal power, where all four blocks reach the same peak. Squares mark the blocks and thin lines are isotherms; panels (b) and (c) share a colour scale.

Table 2: Physical and numerical parameters. The die thickness is an assumption of this study and is not part of the original problem specification.

Quantity	Symbol	Value	Quantity	Symbol	Value
Die size	$L_x \times L_y$	10×10 mm	Grid	$N_x \times N_y$	40×40
Conductivity	k	150 W/(m K)	Node spacing	Δx	0.25 mm
Thickness (assumed)	t_d	0.5 mm	Edge temperature	T_{amb}	300 K
Power budget	$P_{\max}$	5 W	Block centres	(row, col)	(10,10), (10,29), (29,10), (29,29)

Two scenarios were run on the same grid and block centres. In Scenario A the four blocks are single-node point-like sources, the symmetric configuration implied by the problem statement. In Scenario B the blocks are squares of 2×2, 4×4, 6×6 and 8×8 nodes, that is 0.5, 1.0, 1.5 and 2.0 mm on a side, assigned to blocks B1 to B4. Scenario B is our own addition: the footprints are illustrative and were chosen to break the symmetry, because Section 3.4 shows that a symmetric layout leaves nothing for the optimiser to improve. The baseline in both scenarios is the equal split of 1.25 W per block.

5. Results and Discussion

5.1 Numerical verification

Table 3 summarises the checks on the numerical machinery. The matrix $\mathbf{K}$ has 7376 non-zeros out of 1600^2, a fill of 0.29%, which confirms that sparse storage is appropriate, and all 156 boundary rows are strictly diagonally dominant, in agreement with Section 3.3. The condition number of the whole matrix is of order 10^8, but this is an artefact of mixing identity boundary rows (diagonal entry 1) with scaled interior rows (diagonal about 6.4×10^7 m^{-2}). The meaningful measure is the spectral condition number of the interior block, which is only about 616, so the linear solves are well conditioned. The residual of the heat-equation solve is at round-off level, and the full and reduced programs return the same optimum to within 10^{-8} K with HiGHS reporting status 0 (optimal).

Table 3: Numerical diagnostics of the discretisation and the solver.

Diagnostic	Value	Diagnostic	Value
Non-zeros in $\mathbf{K}$ (fill)	7376 (0.29%)	Relative residual (A) / (B)	3×10^{-15} / 1×10^{-14}
Interior eigenvalues, min / max (m^{-2})	2.08×10^5 / 1.28×10^8	$\lvert t^\star_{\text{full}} - t^\star_{\text{red}} \rvert$	$< 10^{-8}$ K
Interior spectral condition number	616	HiGHS status; full LP size	0 (optimal); 1605 variables

5.2 Scenario A: identical point-like blocks

In the symmetric configuration the four blocks have the same self-impedance, 9.009 K W^{-1}. With 1.25 W in each block the peak temperature is 312.55 K, reached at all four block centres, and the linear program returns $q_j^\star = 1.250$ W for every block, identical to the baseline within solver tolerance. This is the outcome predicted by the symmetry argument: when every block is interchangeable no reallocation can lower the maximum, and Fig. 1(a) shows four equal peaks arranged symmetrically about the die centre. Written literally with $\sum_j q_j \le P_{\max}$, the same program returned $\mathbf{q} = \mathbf{0}$ and $t^\star = 0$, which is why the equality constraint is needed. The capstone brief quotes a larger gain for this problem (uniform peak near 343 K against about 327 K), based on moving power from edge blocks to central ones; with the four specified positions, which are mirror images of one another, no such gain exists, and we report this rather than force it.

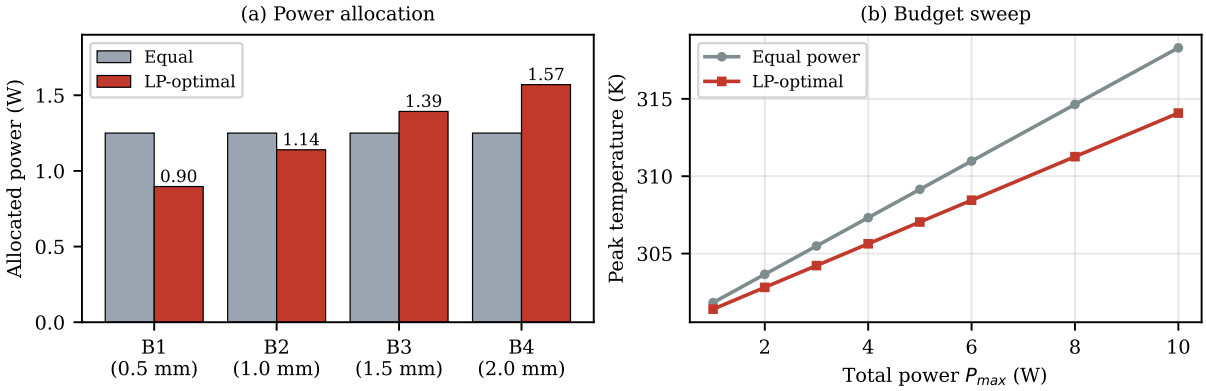


Figure 2: Scenario B. (a) Power per block under equal and LP-optimal allocation, with footprints in brackets. (b) Die peak temperature against total power budget for both allocations.

5.3 Scenario B: blocks with different footprints

When the footprints differ, the same power produces different temperatures. Table 4 shows that one watt in the smallest block B1 raises its local peak by 6.23 K, whereas one watt in the largest block B4 raises its peak by only 3.55 K, because the larger footprint spreads the power over more area and lowers the local power density. Under equal power the block peaks therefore range from 309.15 K for B1 down to 306.02 K for B4, a spread of 3.1 K, and the die peak is set entirely by B1, visible as the hot square at lower left in Fig. 1(b).

Table 4: Scenario B: footprints, self-impedance (peak rise in the block per watt in that block alone), power allocations and block peak temperatures.

	Footprint	Self-impedance	Power (W)		Block peak (K)	
Block	(mm)	(K/W)	Equal	LP	Equal	LP
B1	0.5×0.5	6.233	1.250	0.897	309.149	307.040
B2	1.0×1.0	4.908	1.250	1.140	307.545	307.040
B3	1.5×1.5	4.060	1.250	1.393	306.484	307.040
B4	2.0×2.0	3.552	1.250	1.570	306.022	307.040

The optimiser responds by shifting power from the small, thermally vulnerable block to the large, forgiving ones (Fig. 2a). The optimal split gives B1 0.90 W (17.9% of the budget), B2 1.14 W (22.8%), B3 1.39 W (27.9%) and B4 1.57 W (31.4%), ordered by footprint area. All four blocks then peak at the same 307.04 K, as the identical peak heights in Fig. 1(c) and the last column of Table 4 show. This equalisation is the signature of a minimax optimum: no block can be cooled further without heating another, because any power taken from one block must be placed on a block already at the limit. The die peak falls from 309.15 K to 307.04 K, a reduction of 2.11 K, which is 23.1% of the rise above ambient. The cost is that B4 now runs 1.02 K hotter than before, but the thermal load is more even, which is the desired outcome for an electromigration-limited design.

5.4 Effect of the power budget and grid resolution

Fig. 2(b) and Table 5 show the peak temperature as the budget varies from 1 to 10 W. Both curves are straight lines through ambient. This follows from Eq. (7): $\boldsymbol{\theta} = \mathbf{Zq}$ is linear, and the optimal split depends only on the shape of $\mathbf{Z}$ and not on the scale of the budget, so the peak rise is proportional to $P_{\max}$. The slopes are 1.830 K/W for equal power and 1.408 K/W for the optimal split, so the optimal allocation always removes the same 23.1% of the rise, while its absolute benefit grows from 0.42 K at 1 W to 4.22 K at 10 W. The optimiser is therefore most valuable exactly when the chip is under the highest thermal stress, and because the optimal fractions do not change with $P_{\max}$, the split could be computed once and applied at any power level, provided the model assumptions hold.

Table 5: Scenario B: peak temperature (K) against total power budget.

$P_{\max}$ (W)	1	2	3	4	5	6	8	10
Equal power	301.830	303.660	305.489	307.319	309.149	310.979	314.638	318.298
LP-optimal	301.408	302.816	304.224	305.632	307.040	308.448	311.263	314.079
Reduction	0.422	0.844	1.266	1.687	2.109	2.531	3.375	4.219

Table 6 tests sensitivity to the mesh by refining the grid from 40×40 to 160×160 at fixed die size. In Scenario B the blocks keep their physical size and the results converge: the equal-power peak moves by 0.14 K and the LP optimum by 0.14 K, so the conclusions about the allocation are not artefacts of the grid. In Scenario A the single-node blocks shrink as the mesh is refined and the peak keeps rising, as expected for a point source in two dimensions, whose temperature grows logarithmically as the source size tends to zero. The absolute temperatures

of Scenario A are therefore properties of this mesh and not physical predictions.

Table 6: Peak temperature (K) under grid refinement at fixed die size.

Grid	A: equal power	B: equal power	B: LP-optimal
40×40	312.55	309.149	307.040
80×80	314.60	309.250	307.131
160×160	316.54	309.284	307.182

6. Limitations

The physical model is deliberately simple. It is two-dimensional, treats the die as a thin plate that conducts heat only in the plane, and lets all heat leave through four edges held at exactly 300 K. A real chip loses most of its heat vertically through the substrate, spreader and heat sink, and the package adds finite thermal resistances, so real temperature rises would differ substantially. Silicon conductivity is taken as constant although it falls with temperature, and the analysis is steady-state only, so it says nothing about transient hotspots from bursty workloads.

The absolute temperatures also rest on assumptions that the problem statement does not fix. The die thickness of 0.5 mm was assumed, and because the rise scales inversely with thickness a thinner die would give proportionally larger rises; the 23.1% relative improvement is unaffected but the kelvin values are not. The footprints of Scenario B are illustrative, uniform power density within a block idealises real non-uniform activity, and the point-like blocks of Scenario A are mesh-dependent, as Table 6 shows. No result has been validated against measurements or an established simulator such as HotSpot or a finite-element package, so the verification demonstrates internal consistency and not physical accuracy. Finally, block locations are fixed, so power is optimised but not floorplan, and every block may run at any power from zero upwards, ignoring minimum operating power and performance trade-offs. The link between lower temperature and longer lifetime rests on the electromigration literature [9] and was not modelled.

7. Conclusion

This work combined a conduction model of a silicon die with an exact linear program to find the power split that minimises the peak temperature. The finite-difference Poisson model was verified to round-off residual and good conditioning, the thermal impedance matrix made the problem linear, and the epigraph form let HiGHS return a certified optimum that an independent reduced formulation reproduced. Three findings stand out. A symmetric layout of identical blocks leaves no room for improvement because the equal split is already optimal. When footprints differ, the optimal allocation equalises the block peaks, here lowering the peak by 2.11 K or 23.1% of the rise, and the benefit grows linearly with the budget. And the budget must be imposed as an equality, since the inequality form returns the trivial zero-power solution.

Several extensions would make the model more realistic. A vertical heat path and a finite heat-sink resistance would turn it into a three-dimensional or compact-network problem; temperature-dependent conductivity would make it nonlinear but tractable by iteration; per-block power limits and a performance objective would turn the minimax problem into a speed–temperature trade-off still within the convex framework; time-varying workloads could use the transient impedance $Z_{th}(s)$ of Section 3.2; and block placement could be optimised jointly with power. Validation against a reference thermal simulator or measured data is the most important next step before the numbers inform any design decision.

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